The Likert Blindspot: Detecting Hidden Style Bias in LLM Judges via Controlled Style Decomposition

Anonymous ACL submission

Abstract

Likert scoring—the default evaluation mode for LLM judges—attenuates style bias signals that pairwise comparison detects. We introduce Controlled Style Decomposition (CSD), a black-box framework combining human-validated, NLI-gated content control, per-axis style decomposition, and position-controlled pairwise comparison to quantify style bias in LLM judges. Across four judges from independent providers, one negative control, and three style axes, CSD demonstrates the Likert Blindspot: a severity spectrum from complete attenuation at moderate bias to over 60% signal loss at stronger effects, confirmed by seven statistical methods across three benchmarks. GEE conditional decomposition identifies a per-axis confound structure consistent across all four target judges: verbosity bias is style-dominated, formality exhibits a conditional effect moderated by content quality, and register is judge-dependent—determining which estimates are directly interpretable and which require conditional analysis. A negative-control judge shows no detectable bias on either primary axis, supporting discriminative validity. Whether the Blindspot reflects measurement attenuation or genuine construct divergence between absolute and comparative evaluation remains an open question; under either interpretation, Likert-only evaluation underestimates style-driven preferences that propagate through pairwise-based alignment pipelines.

1 Introduction

"What is the average height for a male?" — given a formal and a casual answer with verified identical content (NLI entailment = 0.993), GPT-4o rates the casual variant *higher* on a 1–10 Likert scale (8 vs. 5), yet consistently selects the formal variant in pairwise comparison. Same judge, same texts, opposite conclusions. This is the *Likert Blindspot*. We note that formality preferences may partially reflect legitimate quality judgments—our GEE decomposition (§5.2) quantifies this originality component. This example reflects a systematic pattern.

LLM judges are central infrastructure for NLP evaluation and alignment training (RLHF, DPO), where judge preferences serve as reward signals; any systematic bias propagates through both evaluation leaderboards and model development pipelines. Style features alone explain most of the variance in judge scores (Murugadoss et al., 2025): 76–92% of observed preferences originate from surface-level style cues (Li et al., 2026), with bias scores of 0.76–0.92 across five judges (Soumik et al., 2026). Despite this consensus, existing approaches address individual requirements (multi-dimensional style analysis (Cho et al., 2026; Yang et al., 2026), interventional design (Reber et al., 2025), population-level correction (Dubois et al., 2025)), but no single framework integrates per-axis decomposition, interventional identification, per-instance diagnostics, and black-box applicability. A further concern: standard Likert scoring produces ∼60% uninformative ties, and even non-tied pairs carry weaker signal than pairwise comparison—a limitation predicted by measurement theory (Thurstone, 1927) but previously unquantified for LLM judges.

We propose Controlled Style Decomposition (CSD), combining NLI-gated content control, per-axis style decomposition, and position-controlled pairwise comparison. Inspired by interventional text analysis (Veitch et al., 2020), CSD operates in three stages: style-controlled rewriting along a single axis, NLI entailment verification, and bias measurement via forced-choice comparison. CSD requires no access to judge internals and applies post-hoc to any black-box judge. Our experiments demonstrate the *Likert Blindspot* across four target judges from independent providers (GPT-4o, Qwen3-32B, Llama-3.3-70B, Gemini-2.5-Flash), one negative control (Claude Sonnet 4), three benchmarks, and two primary style axes, with ex-

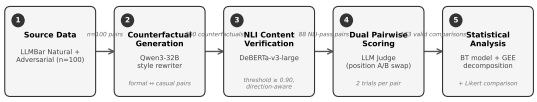


Figure 1: CSD pipeline overview. Numbers indicate sample attrition at each stage.

ploratory register evidence. On formality data, Likert yields non-significant ATE ($p = 0.332$, 0.908), while BT detects 63.0% (GPT-4o, $p = 0.0004$) and 66.5% (Qwen3-32B, $p < 0.0001$) formal preferences; verbosity shows 77.3% BT ($p < 0.0001$) invisible to Likert ($p = 0.179$; Figure 3).

Our contributions are both methodological and empirical:

- **CSD**, a black-box framework for per-axis style bias detection via NLI-gated content control, style-controlled rewriting, and position-controlled pairwise comparison (§3).
- **The Likert Blindspot:** Likert scoring attenuates style bias signals that pairwise comparison detects—complete at moderate bias ($d \to 0$ at $n = 355$), >60% signal loss at stronger effects—across six judge×axis configurations ($p < 0.001$ pairwise vs. $p > 0.17$ Likert), confirmed by seven statistical methods (Tables 1–2).
- **Per-axis confound decomposition:** GEE conditional analysis separates style from quality effects, showing verbosity is style-dominated (Style OR 8–13, Orig OR <1), formality conditional, register judge-dependent—determining which estimates are directly interpretable and which require conditional analysis (Table 3).

2 Related Work

Two lines of work motivate our approach: interventional methods for diagnosing judge bias, and measurement theory on the sensitivity of different elicitation formats.

Interventional Approaches to Judge Bias Text-based interventional methods have a growing methodological foundation (Feder et al., 2021; Wu et al., 2021; Feder et al., 2022). Reber et al. (2025) propose RATE, a framework that rewrites model outputs along style attributes using imperfect counterfactuals and measures the causal effect on reward model scores via Likert-scale Average Treatment Effect (ATE). RATE applies a correction term to account for rewriting imperfections, but its reliance on Likert ATE as the primary metric is a vulnerability: our experiments show that Likert scoring exhibits a ceiling effect in which 60% of style-transformed pairs receive identical integer scores, yielding non-significant ATE estimates ($p = 0.332$) for the same interventions that pairwise comparison detects at $p = 0.006$. If this Likert Blindspot generalizes beyond our setting, RATE's own style-related findings may underestimate the true magnitude of attribute effects. CSD differs from RATE in three respects: it targets individual style axes rather than general attributes, it measures bias through position-controlled pairwise comparison rather than Likert ATE, and it enforces content preservation via NLI entailment gating rather than post-hoc correction for imperfect rewrites. Qiu et al. (2025) also adopt an interventional LLM-based design, rewriting texts to measure rhetorical style effects and aggregating preferences with a Bradley-Terry model. Their framework operates at persona-level granularity (e.g., "writes like a professor"), whereas CSD intervenes on specific style axes (e.g., formality, conciseness) and uses a sign test with position control rather than parametric aggregation.

Measurement Sensitivity in NLG Evaluation The distinction between ordinal and interval measurement dates to Stevens (1946), whose taxonomy of scales remains foundational. The superiority of pairwise comparison over rating scales for eliciting reliable human judgments has been established since Thurstone (1927), whose Law of Comparative Judgment formalizes why forced-choice comparisons yield finer discriminability than absolute ratings. Preston and Colman (2000) and Simms et al. (2019) show that scale granularity affects reliability and psychometric precision, while Krosnick (1999) documents satisficing behavior that may compound the tie inflation we observe. DeCoster et al. (2011) show that artificial categorization of continuous variables discards variance and reduces statistical power, directly relevant to the Likert Blindspot mechanism. Within NLP, Amidei et al. (2019) review the use of rating and Likert scales in NLG evaluation, identifying inconsistencies in scale design and reporting

that compromise cross-study comparability. Clark et al. (2021) demonstrate systematic flaws in human evaluation; Celikyilmaz et al. (2020) survey evaluation methods, and van der Lee et al. (2019, 2021) establish best practices for human NLG evaluation. Kiritchenko and Mohammad (2017) and Novikova et al. (2018) confirm that forced-choice methods (best-worst scaling, ranking) detect quality differences that rating scales miss; Louviere et al. (2015) provide the theoretical foundation. In machine translation, Bojar et al. (2016) document the practical tension between pairwise ranking and direct assessment at scale—WMT's transition to rating-based evaluation was driven by quadratic cost, not by measurement superiority. Our contribution is the first controlled quantification of this per-axis sensitivity gap in the LLM judge setting, identifying the two-layer mechanism (tie inflation plus residual signal attenuation) that drives it.

Style Bias Detection and Mitigation The foundational LLM-as-judge framework was established by Zheng et al. (2023), who introduced MT-Bench and Chatbot Arena and documented position bias in pairwise evaluation. Wang et al. (2024) further showed that preference rankings can be manipulated by swapping option order, proposing a win/tie/lose annotation scheme, and Shi et al. (2025) systematically study position bias across 15 judges, showing it varies by model and quality gap. Li et al. (2025) address selection bias through calibration of prediction distributions during inference. These works focus on *position* bias and calibration; Ye et al. (2025) systematically quantify 12 bias types using automated counterfactual modifications, and Koo et al. (2024) benchmark cognitive biases in LLM evaluators, though neither provides per-axis isolation of individual style effects. Style bias as a separate, dominant bias source has received attention only recently (Wu and Aji, 2025; Murugadoss et al., 2025; Li et al., 2026; Soumik et al., 2026); Saito et al. (2023) specifically quantify verbosity bias in multi-turn chatbot evaluation, and Chen et al. (2024) compare human and LLM judge biases, finding partially overlapping but distinct bias profiles; Bavaresco et al. (2025) show that LLM–human agreement varies substantially across tasks. Li et al. (2026) further show that style-mediated *preference leakage* propagates through the evaluation pipeline, motivating per-axis isolation of style effects. Tan et al. (2025) provide JudgeBench, a benchmark revealing that even strong judges perform near chance on challenging pairs, underscoring the need for controlled experimental designs rather than aggregate accuracy metrics. Multi-feature frameworks such as CASSE (Cho et al., 2026) and FairJudge (Yang et al., 2026) decompose style into finer dimensions but rely on correlational analyses—they cannot distinguish whether observed associations reflect causal style effects or confounds with content quality, which our NLI-gated rewriting design addresses. Stureborg et al. (2024) demonstrate inconsistency and bias across evaluation formats, and Park et al. (2024) show that verbosity and other surface features systematically inflate judge scores. Crucially, Jeong et al. (2025) identify the Comparative Trap: pairwise comparison amplifies style biases relative to pointwise scoring, raising the concern that our pairwise-based detection may partly reflect format-level amplification rather than pure detection. We address this directly in §5.4.

Bradley-Terry Models in Evaluation The Bradley-Terry model (Bradley and Terry, 1952) underpins Chatbot Arena (Zheng et al., 2023; Chiang et al., 2024) and AlpacaFarm (Dubois et al., 2023); Boubdir et al. (2024) establish best practices, and Liusie et al. (2024) show zero-shot LLM pairwise comparison matches pointwise evaluation. We adopt BT because it uses all comparison data including inconsistent pairs, yielding proper effect-size CIs; our contribution is to measure the Likert–pairwise sensitivity gap for style bias and show that per-axis GEE decomposition exposes confound structures invisible to aggregate scores.

3 Method

3.1 Style-Controlled Rewriting

We isolate the effect of individual style attributes on judge scores by generating text pairs that differ along exactly one style axis. For each text, an instruction-tuned LLM (Qwen3-32B) classifies its current style along the target axis (e.g., formal vs. casual via binary classification). It then rewrites the text toward the opposite pole (temperature 0.7) according to an axis-specific prompt (e.g., "Rewrite the following text to be more formal/casual while preserving the exact same content and meaning"). This produces a dataset balanced across both directions at the dataset level: texts detected as formal are rewritten to casual, and vice versa. Each operation constitutes a binary intervention on a single style variable while all other axes remain fixed

(Section 3.5). To control for rewriting artifact bias—the possibility that the judge prefers original text over any rewrite regardless of style—we additionally generate a same-style double-rewrite for each text (e.g., formal → formal) to form an original-vs-double-rewrite control pair, following the RATE bias correction protocol (Reber et al., 2025).

3.2 Content Preservation Verification

Style-controlled rewriting risks altering semantic content, which would confound the style effect with a content quality effect (the $R \to Q$ path in Figure 2). To block this path, we apply a natural language inference (NLI) gate using DeBERTa-v3-large with an entailment threshold of 0.90, selected as the threshold above which NLI performance plateaus (Appendix E). For axes that preserve information density (formality, register), we apply standard bidirectional entailment: both NLI(original → rewrite) and NLI(rewrite → original) must exceed this value. For axes that inherently change information density (e.g., verbosity), bidirectional verification is inappropriate—expansion necessarily introduces new content that the original cannot entail. We instead apply direction-aware verification: for expansions, only backward entailment (rewrite entails original); for compressions, only forward entailment (original entails rewrite). This recovered verbosity from 65% to 86% pass rate without inflating the effect estimate; for formality, strict bidirectional gating is necessary (Appendix F). This gating mechanism provides filtering-based confounder control (Veitch et al., 2020) by ensuring that retained pairs differ only in style, not in content quality (validated by human annotation; §5.4). BERTScore serves as a supplementary monitor but does not participate in gating.

3.3 Measurement Protocol

Each pair that passes the NLI gate is evaluated by four target judges—GPT-4o (`gpt-4o-2024-08-06`), Qwen3-32B, Llama-3.3-70B, and Gemini-2.5-Flash—under two complementary scoring modes. In Likert mode, the judge assigns an integer score from 1 to 10 for overall quality to each text independently. In pairwise mode, the judge receives both texts as candidates A and B and makes a forced choice of which response is better.

To control for position bias, each pairwise comparison is evaluated twice with the presentation order swapped, which produces $n_{\text{pairs}} \times 2$ observations. We apply two analysis strategies to these data. For the nonparametric sign test, we retain only *consistent* pairs where the judge selected the same text in both orderings. For the Bradley-Terry model, we use all observations including inconsistent pairs, as the model naturally accommodates noise in individual comparisons.

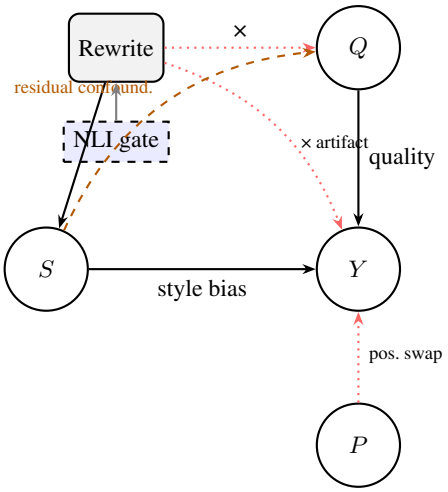


Figure 2: Assumed data-generating process (design rationale, not a claim of perfect identification). The rewriting mechanism R targets style S but may inadvertently alter content quality Q (dotted red, blocked by NLI gate) or introduce rewriting artifacts ($R \to Y$, blocked by double-rewrite control). Position bias $P \to Y$ is controlled by order swapping. The dashed orange $S \to Q$ path represents residual confounding: style may influence perceived quality through channels the NLI gate cannot block (Limitations). Y: judge score.

3.4 Experimental Setup

We evaluate on LLMBar (Zeng et al., 2024), a benchmark of $n = 100$ instruction-response pairs spanning diverse task types, designed to test LLM judges' ability to distinguish response quality with human-verified quality labels. LLMBar is well-suited for style bias detection because its human-verified labels provide ground truth against which judge deviations can be measured, and its moderate size permits exhaustive pairwise evaluation with position control. We select formality and verbosity as primary axes based on prior evidence that these are among the most commonly reported style biases in LLM judges (Wu and Aji, 2025; Saito et al., 2023).

3.5 Design Rationale

Figure 2 presents the assumed data-generating process. Three variables govern the observed judge

score Y: the style attribute S (e.g., formality), the original content quality Q, and the judge's internal scoring function. Both S and Q causally influence Y; the goal of CSD is to approximate the effect $S \to Y$ while blocking confounding through Q. Style-controlled rewriting generates a text pair that differs along a single style axis, but any rewriting operation risks inadvertently altering content quality ($R \to Q$). The NLI gate (§3.2) blocks this path, yielding a controlled estimate of the $S \to Y$ effect. Our approach provides interventional rather than formal causal evidence: we do not claim that this DAG is a complete structural causal model, nor do we invoke SUTVA or consistency assumptions from the potential outcomes framework. "Controlled" in this context refers to experimentally controlled rewriting conditions, not formal causal identification in the potential-outcomes sense. Our identification rests on three operational assumptions: (i) the NLI gate approximates content preservation, blocking the $R \to Q$ path; (ii) position swapping controls for presentation-order effects; and (iii) the style rewrite introduces no systematic differences beyond the target axis.

Table 1: Main results across five judges, three style axes, and three benchmarks. Each row tests one judge–axis pair: BT > 0.50 indicates style preference; † marks Blindspot cases (BT significant, Likert not). BT: Bradley-Terry win probability (95% CIs in Appendix I). AEval: AlpacaEval; MTB: MT-Bench.

Bench	Judge	Axis	BT P	Likert d (p)
LLMBar	GPT-4o	Form.	.630***	.102 (.332)
	GPT-4o	Verb.	.760***	−.095 (.403)
	GPT-4o	Reg.	.570†	−.031 (.785)
	Qwen3	Form.	.665***	.011 (.908)
	Qwen3	Form.^{b}	.675***	−.025 (.600)
	Qwen3	Verb.	.773***	−.151 (.179)
	Qwen3	Reg.	.553	−.073 (.505)
	Llama-70B	Form.	.695***	.010 (.846)
	Llama-70B	Verb.	.735***	.020 (.752)
	Llama-70B	Reg.	.635†	.024 (.690)
	Gemini	Form.	.574*	−.082 (.309)
	Gemini	Verb.	.708***	.051 (.370)
	Claude	Form.	.562	−.021 (.721)
	Claude	Verb.	.541	−.039 (.743)
AEval	GPT-4o	Form.	.727***	.148 (.170)
	Qwen3	Form.	.657***	.081 (.198)
	Qwen3	Verb.	.841***	.375 (.0003)
MTB	GPT-4o	Form.	.692***	.167 (.016)
	Qwen3	Form.	.712***	.149 (.007)

*** $p < 0.001$; * $p < 0.05$ (Holm-adj. for primary; raw for $\dagger_c$).
†: Blindspot; ‡: both sig.; —: neither; $\dagger_e$: expl.; $\dagger_c$: confirm.
^{b} Scaled replication ($n = 355$); other LLMBar $n = 88$. Negative d: styled vari
Empty cells: untested benchmark–axis combinations; cross-bench. validation ta

4 Experiments

4.1 Statistical Framework

The primary family comprises six tests—three target judges (GPT-4o, Qwen3-32B, Llama-3.3-70B) $\times$ two axes (formality, verbosity)—corrected via Holm-Bonferroni at $\alpha = 0.05$; Gemini-2.5-Flash serves as confirmatory replication and is excluded from FWER control; as a sensitivity check, all eight tests remain significant under the expanded family (Appendix Q). Table 1 reports both raw and adjusted p-values. Analyses outside this grid are classified as either *exploratory* (NLI threshold sensitivity sweep, §5.4) or *confirmatory* (consistency filter validation, §B). Exploratory analyses are reported with uncorrected p-values and interpreted as hypothesis-generating; confirmatory analyses test specific hypotheses about pipeline validity and are evaluated at the nominal $\alpha = 0.05$ level without additional correction, as they address orthogonal questions about measurement reliability rather than the primary style-bias family. Bradley-Terry parameters are estimated via spectral initialization followed by minorization-maximization (Hunter, 2004), using the default regularization of the `choix` Python library.[1] All bootstrap confidence intervals use 10,000 pair-level resamples with seed 42. GEE models use exchangeable working correlation, justified by near-zero estimated within-pair correlations ($\hat{\alpha} \in [-0.25, +0.11]$; Appendix C).

Model selection. Target judges span four architectures: GPT-4o (2024-08-06), the dominant judge powering AlpacaEval and MT-Bench leaderboards; Qwen3-32B-Instruct, also serving as rewriter for methodological consistency; Llama-3.3-70B-Instruct, the latest dense Llama model (Llama 4's MoE architecture introduces an additional confound); and Gemini-2.5-Flash as a confirmatory fourth architecture. Claude Sonnet 4 serves as negative control. We pin versions for reproducibility; CSD applies to any black-box judge.

[1] In binary-choice BT models, IIA is structurally satisfied; the direction asymmetry in Appendix H (formality: formal-origin 88.9% vs. casual-origin 52.6%, Fisher $p = 0.015$; verbosity: $p = 0.453$) reflects heterogeneous treatment effects across style directions rather than IIA violation.

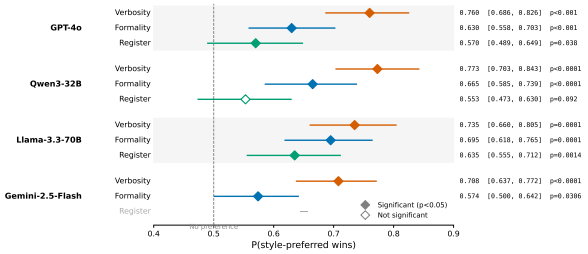


Figure 3: Bradley-Terry win probabilities across four target judges and three style axes. Filled diamonds: significant ($p < 0.05$); open diamonds: non-significant. Error bars: 95% bootstrap CI. The dashed line at 0.50 indicates no style preference. Verbosity shows the strongest and most consistent bias across all four judges.

5 Analysis

5.1 The Likert Blindspot

Head-to-head comparison with RATE. Applying RATE's Likert ATE methodology (Reber et al., 2025) to our formality style-transformed pairs yields Cohen's $d = 0.10$, $p = 0.33$ (non-significant, $n = 88$); pairwise comparison on identical data reveals a 63.0% formal preference ($p = 0.0004$, Bradley-Terry on all 173 comparisons), confirmed by a consistency-filtered sign test (73.9%, $p = 0.006$, $n = 46$ consistent pairs). This is the Likert Blindspot: the same data, the same style intervention, two measurement instruments—one detects bias, the other does not. A $4\times$ scaled replication ($n = 355$) distinguishes measurement attenuation from insufficient power: the Likert effect size shrinks to $d = -0.025$ (near zero) while BT remains stable at 67.5% ($p < 0.0001$)—if the gap were merely statistical power, d would remain non-zero at larger n. Bootstrap subsampling confirms: d remains flat near zero ($d_{\text{median}} = -0.017$) as CIs narrow from ± 0.3 to ± 0.1, never excluding zero (Appendix J). Attenuation follows a severity spectrum: at moderate bias (BT 63–67%), it is complete ($d \to 0$ at $n = 355$); at stronger effects (BT >70%), Likert partially detects but attenuates >60% (MT-Bench $d \approx 0.15$ for BT ∼70%)—explaining why three of eleven bias-positive cells escape the Blindspot (Table 1, ‡ rows).

The frameworks are complementary: RATE measures aggregate quality via Likert ATE; CSD targets per-axis style preference via forced-choice comparison, eliminating ties. The Blindspot arises because Likert tie rates cluster at 58–73% across all nine configurations; a 100-point continuous scale does not resolve this ($p = 0.725/0.117$; Appendix M), confirming a paradigm difference rather than granularity limitation.

Table 2: Sensitivity hierarchy: seven measurement methods applied to the same formality style-transformed data. Only pairwise methods (top two rows) reach significance. All Likert-based methods fail.

Method	Statistic	p	Data
BT model	63.0% [55.7, 70.3]	0.0004	All 173 comp.
Sign test (filtered)	73.9% (34/46)	0.006	Consistent
Likert ATE (t-test)	$d = 0.102$	0.332	NLI≥0.90
Binarized Likert	60.6%	0.296	Non-tied
Ordinal reg. (prop. odds)	$\beta = -0.079$	0.769	NLI≥0.90
GEE (exchangeable)	$\beta = -0.084$	0.587	NLI≥0.90
Wilcoxon signed-rank	$W = 274$	0.905	NLI≥0.90

Sensitivity hierarchy across measurement methods. To confirm that the Likert Blindspot is not an artifact of a particular statistical test, we apply seven measurement methods to the same formality data and rank them by sensitivity (Table 2). Both pairwise methods detect the formality preference; all Likert-based methods fail regardless of whether they assume interval scaling (ATE t-test), ordinal scaling (ordinal logistic regression), or non-parametric ranking (Wilcoxon signed-rank). The same attenuation pattern replicates on verbosity (Appendix L), where all six Likert methods remain non-significant despite a stronger pairwise signal (BT 77.3%, $p < 0.001$).

Figure 4 plots detection probability at observed effect magnitudes (not post-hoc power; cf. Hoenig and Heisey 2001): across six cells, Likert requires 4–206$\times$ the current n (336–27,159 pairs; Appendix K) to detect effects BT identifies at $p < 0.001$ with $n \leq 100$.

Measurement attenuation vs. construct divergence. An alternative interpretation is that Likert and pairwise modes measure different constructs—absolute quality vs. ordinal preference—making the divergence a construct difference rather than measurement failure. Tripathi et al. (2025) independently confirm that pairwise and pointwise protocols exhibit distinct sensitivity profiles, with pairwise preferences being more volatile—consistent with this construct divergence interpretation. Both interpretations lead to the same conclusion: Likert audits miss style preferences that drive pairwise-based alignment pipelines (RLHF, DPO). The Blindspot manifests in eight of eleven bias-positive cells; three exceptions (BT >69%, $d \geq 0.14$) confirm it is effect-size dependent.

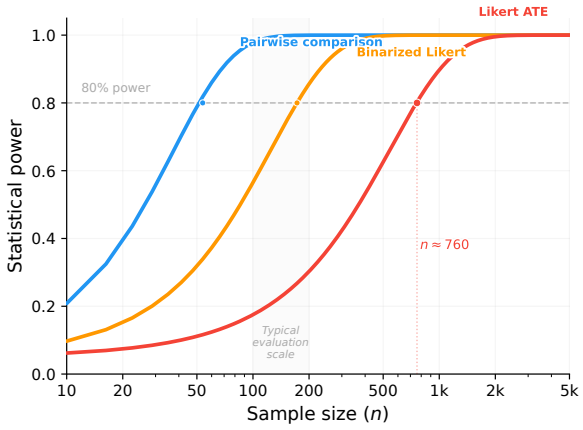


Figure 4: Detection probability at observed effect magnitudes. 80% power crossings: sign test $n \approx 62$, BT $n \approx 114$, binarized Likert $n \approx 462$, Likert ATE $n \approx 1{,}473$. At typical scales ($n = 50$–200, shaded), pairwise methods reach 80% power while Likert methods remain far below. At $n = 355$, Likert $d = -0.025$ confirms the gap is structural, not merely a power issue.

Table 3: GEE conditional decomposition (exchangeable correlation) of style and originality effects. Verbosity is style-dominated across all four target judges; formality shows varying confound structure.

	Style OR		Originality OR	
Config	OR	p	OR	p
Qwen3 verb.	13.10	<.0001	0.49	.096
GPT-4o verb.	11.34	<.0001	0.45	.062
Llama-70B verb.	8.21	<.0001	0.38	.020
Gemini verb.	8.98	<.0001	0.46	.061
GPT-4o form.	2.58	.002	2.32	.006
Qwen3 form.	4.20	.0002	4.70	.0001
Qwen3 form.[a]	2.15	<.0001	4.99	<.0001
Llama-70B form.	4.81	<.0001	3.12	.0007
Gemini form.	1.64	.101	2.51	.002
GPT-4o reg.	1.88	.029	2.21	.006
Qwen3 reg.	1.91	.046	5.72	<.0001
Llama-70B reg.	3.33	.0003	3.07	.0008

[a] $n = 355$; other rows: $n = 88$. 95% CIs in Appendix.

5.2 Per-Axis Confound Structure

GEE conditional decomposition (Liang and Zeger, 1986) with exchangeable working correlation and robust standard errors separates style from originality effects (Appendix A; model diagnostics including correlation structure robustness and interaction tests in Appendix C). After controlling for original preference, the formality effect remains significant (GPT-4o: OR $= 2.58$, $p = 0.002$; Qwen3-32B: OR $= 2.15$, $p < 0.0001$, $n = 355$; Llama-70B: OR $= 4.81$, $p < 0.0001$; Table 3); the significant originality ORs confirm residual confounding.

The confound structure is consistent across all four target judges for verbosity (style-dominated: Style OR 8.2–13.1, Orig OR 0.38–0.49); formality is conditionally style-driven in three judges but originality-dominated in Gemini; register is judge-dependent. Verbosity Style ORs of 8–13 are comparable to the strongest epidemiological effect sizes, indicating a dominant driver of pairwise outcomes rather than a subtle confound. Same-style double-rewrite controls confirm a rewriting artifact (Appendix O), absorbed by GEE's originality term.

5.3 Cross-Judge and Cross-Axis Replication

Three independent judges replicate the formality Blindspot: Qwen3-32B (BT 66.5%, $p < 0.0001$; Likert $p = 0.908$), Llama-3.3-70B (BT 69.5%, $p = 0.0001$; Likert $p = 0.846$), and Gemini-2.5-Flash (confirmatory: BT 70.8%, $p < 0.0001$; Likert $d = 0.05$, n.s.). Gemini's formality confound is originality-dominated (Style OR $= 1.64$, n.s.; Orig OR $= 2.51$, $p = 0.002$), unlike the other three judges (§5.2).

Verbosity provides the strongest Blindspot: all four target judges show consistent preference (GPT-4o 76.0%, Qwen3-32B 77.3%, Llama-70B 73.5%, Gemini 70.8%; all $p \leq 0.0001$), while Likert ATE remains non-significant for all ($p = 0.179$–0.752). GEE confirms massive style effects (OR 8.2–13.1, all $p < .0001$) with originality ORs below 1 (0.38–0.49; only Llama-70B significant at $p = .020$, yet Style OR exceeds Orig OR by 22×), confirming style dominance (Table 3). Both rewriting directions favor verbose responses (90.7%/80.6%); controlling for length leaves the style OR unchanged (<2%). Evaluator-producer overlap applies only to Qwen3-32B; an independent GPT-4o rewriter bounds this to a 22% Style OR reduction (13.10 → 10.17; Appendix D).

Across both primary axes, the Blindspot replicates in all four target judges with consistent confound structure: verbosity is uniformly style-dominated while formality is conditionally style-driven—a systematic measurement gap rather than a judge- or axis-specific artifact.

5.4 Robustness

Content preservation. Two independent annotators evaluated 50 NLI-scored pairs (25 gate-passed, 25 gate-rejected). Annotator 1 judged all 25 gate-passed pairs content-preserving (precision 100%; Wilson CI [86.7%, 100%]); Annotator 2 judged 21/25 (84%; CI [65.3%, 93.6%]), with all four dis-

agreements on verbosity-axis condensation pairs. Under majority vote, pass precision is 100% (inter-annotator $\kappa = 0.44$, AC1 $= 0.65$; Appendix S). Annotator 1's disagreements with the NLI gate occur exclusively on gate-rejected pairs; Annotator 2's four gate-passed disagreements all involve verbosity condensation where subjective meaning boundaries differ. This error profile ensures that any detected style bias cannot be attributed to content contamination in admitted pairs. The gate's estimated recall is 59.5% (false-negative rate 40.5%), reflecting conservative calibration toward high precision at the expense of coverage. An independent LLM cross-validation on the full dataset yields consistent results (LLM–NLI AC1 $= 0.89$). BT remains stable ($\leq$3pp variation) across 11 NLI thresholds (Appendix E), and no covariate differences exist between retained and discarded pairs ($p > 0.12$; §B).

Detection vs. amplification. Jeong et al. (2025) show that pairwise comparison amplifies style biases (the Comparative Trap). The divergence between Likert and pairwise signals—whether reflecting Likert attenuation, pairwise amplification, or construct-level differences—cannot be definitively resolved without human ground-truth bias measurements; six independent controls bound the amplification risk: (i) NLI gate ablation reverses NLI-fail pairs to BT 41.7% (Appendix N); (ii) Claude Sonnet 4 shows no bias with the identical pairwise format (BT 56.2%/54.1%, n.s.; TOST equivalence at $\Delta = 15$pp, the narrowest margin achieving adequate power at $n = 87$; $\Delta_{\min} = 15.7$pp; Appendix G); (iii) cross-axis BT variation (register 55% $<$ formality 67% $<$ verbosity 77%) is inconsistent with uniform format amplification; (iv) rewriting artifact absorbed by GEE decomposition (Appendix O); (v) debiased prompting reduces bias with axis-dependent effectiveness (verbosity -13.1pp; formality -5.7/-0.4pp), suggesting varying structural depth; (vi) four independent architectures all confirm the Blindspot with consistent GEE confound structure (§5.3). Why Claude does not exhibit the Blindspot remains an open question; possible explanations include differences in RLHF calibration, model scale, or evaluation-specific fine-tuning, but our data do not distinguish among these hypotheses. Register is exploratory and judge-dependent (Llama-70B BT 63.5%, GPT-4o 57.0%, Qwen3-32B n.s.; Table 3).

6 Conclusion

CSD yields two findings. First, the *Likert Blindspot*: our evidence suggests that Likert scoring attenuates style bias signals that pairwise comparison detects—complete at moderate bias ($d \to 0$ at $n = 355$), >60% signal loss at stronger effects—confirmed by seven statistical methods; we cannot fully separate measurement attenuation from format-level amplification (§5.4) or construct divergence, but under any interpretation, Likert-only audits underestimate style-driven preferences that propagate through alignment pipelines. Second, GEE conditional decomposition identifies per-axis confound structure—verbosity style-dominated, formality conditional, register judge-dependent—determining which estimates require conditional analysis.

Existing Likert-based null findings may reflect measurement insensitivity rather than absence of bias; pairwise comparison with amplification controls (§5.4) offers superior detection sensitivity. Under the six amplification controls reported in §5.4—including a negative-control judge that shows no pairwise bias—the observed sensitivity gap is more consistent with measurement attenuation than format-induced amplification, though the latter cannot be fully excluded.

For practitioners auditing LLM judges, our results suggest that Likert-only evaluation is insufficient: bias audits should employ pairwise comparison with position control and per-axis decomposition to distinguish style-dominated from quality-confounded effects. Human ground-truth validation of both measurement modes on identical pairs—determining whether the Blindspot reflects measurement attenuation, construct divergence, or a combination—remains the most informative next step.

Limitations

Quality–style confounding and construct validity. The NLI gate blocks content changes but cannot block quality perception shifts mediated by style itself—a fundamental identifiability limit of text-rewriting approaches. Although the formality effect survives logistic adjustment for original preference (OR $= 2.32$, $p = 0.006$; Appendix A), residual confounding is not fully eliminated. An alternative interpretation of the Blindspot is that Likert and pairwise modes measure different constructs: Likert captures absolute quality difference

while pairwise captures ordinal preference. Under this view, the divergence reflects construct difference rather than measurement failure. The Comparative Trap (Jeong et al., 2025) lends support to this alternative; six experimental controls bound this risk (§5.4), though definitive separation would require human ground-truth ratings. The primary open question is whether pairwise comparison detects genuine style bias or amplifies micro-preferences inherent in the format. Definitive resolution requires human evaluation comparing Likert and pairwise judgments on identical pairs—an important direction for future work. While our controls bound the amplification factor, we cannot quantify it precisely without a ground-truth bias magnitude—an inherent limitation of observational-format comparisons.

Evaluator-producer bias and generalizability. Qwen3-32B serves as both rewriter and judge, raising self-preference concerns (Panickssery et al., 2024); an independent GPT-4o rewriter bounds this to a 22% Style OR reduction (Appendix D). GPT-4o, Llama-3.3-70B, and Gemini-2.5-Flash use a separate rewriter and are not subject to evaluator-producer overlap; all three independently confirm the Blindspot across four architectures, though cross-judge effect-size comparisons are not valid. Cross-rewriter validation shows that Blindspot existence is robust to rewriter identity, but effect-size precision varies by approximately $\pm 22\%$ with rewriter choice (Appendix D). Register results carry the highest evaluator-producer risk—all register rewrites use the same Qwen3-32B rewriter without the cross-rewriter validation applied to formality and verbosity—and register conclusions remain tentative pending independent rewriter validation.

Scope and ecological validity. Primary analyses use LLMBar ($n = 100$); additional axes, larger benchmarks, and further judges are needed. A fourth axis—tone (assertive vs. hedging)—was explored but abandoned: epistemic commitment changes caused the NLI gate to correctly reject most counterfactuals (pass rate $<50\%$), confirming that the quality gate prevents content-altering manipulations from entering the pipeline. Our per-axis decomposition treats each style axis independently; in practice, axes may interact (e.g., formality shifts may co-vary with register), and such confounds are not isolated by the current design. Cross-axis contamination analysis (Appendix R) confirms that formality rewrites produce negligible changes in sentence structure ($|d| < 0.03$) and verbosity rewrites leave lexical diversity unchanged (TTR $|d| < 0.04$). Our empirical evidence covers three style axes (formality, verbosity, register); whether the per-axis confound taxonomy generalizes to other axes (e.g., hedging level, bullet-point vs. prose) remains open. Our controlled manipulation design ensures internal validity but may not fully generalize to naturalistic evaluation settings where style and content covary.

Validation methodology and scale design. NLI validation uses Qwen3-32B as annotator (GPT-4o cross-validation on all 200 pairs confirms 92.1% formality / 82.6% verbosity pass precision); two independent annotators confirm 84–100% pass precision on 50 pairs (§5.4; majority-vote precision 100%), though larger-scale annotation across all axes would further strengthen generalizability. A 100-point continuous scale does not resolve the Blindspot (Appendix M), suggesting the gap reflects measurement mode rather than granularity (Preston and Colman, 2000); the 100-point experiment was tested with a single judge, and multi-judge replication would strengthen this conclusion. Equivalence testing (Lakens, 2017) could formalize whether the Likert effect size is practically zero.

Code and Data

Code and data will be released upon acceptance. *Organization: confound analysis (A–B), model diagnostics (C), independence tests (D), gate validation (E–F), equivalence testing (G), statistical details (H–K), sensitivity analyses (L–N), artifact controls (O), implementation (P), multiplicity (Q), cross-axis contamination (R), human annotation agreement (S).*

A Direction Asymmetry and Quality Confounding

The formality axis exhibits a pronounced direction asymmetry: the judge selects the formal variant in 78.8% of formal-is-rewrite pairs ($p = 0.001$) but only 54.5% of casual-is-rewrite pairs ($p = 0.83$). This asymmetry suggests that original quality Q and formality S are partially confounded—formally written texts may be inherently perceived as higher quality. To quantify this, we fit a GEE logistic model (exchangeable correlation) with both formality and original-preference as predictors. After controlling for original preference, the formality effect remains significant (OR $= 2.58$, $p = 0.002$),

and the original-preference effect is also significant (OR $= 2.32$, $p = 0.006$). The formality preference thus survives adjustment for original-preference confounding, though the significant original-preference effect indicates that quality–style confounding is present and non-negligible.

At $n = 88$, the Qwen3-32B formality Style OR (4.20) exceeded the GPT-4o estimate (2.58), but a scaled replication ($n = 355$) corrects this to 2.15 ($p < 0.0001$), comparable to GPT-4o and consistent with small-sample upward bias rather than evaluator-producer inflation of the style effect. The originality confound remains stronger (4.99 vs. 2.32), consistent with evaluator-producer bias inflating the originality component (Limitations). The directional agreement between judges—both favor formal text after controlling for originality—supports the existence of formality bias, but the effect magnitude from Qwen3-32B should not be directly compared with GPT-4o.

The NLI gate blocks *content* changes but cannot block quality perception shifts that are themselves mediated by style—a fundamental identifiability limit of any text-rewriting approach.

B Consistency Filter Robustness

The position-consistency filter retains 55 of 98 valid pairs (56.1%); consistent and inconsistent groups show no significant differences across four covariates (NLI score $p = 0.824$, BERTScore $p = 0.12$, text length $p = 0.137$, length diff $p = 0.715$), indicating selection on judge decisiveness rather than text properties. The BT model using all 173 comparisons including inconsistent pairs yields 63.0% ($p = 0.0004$), consistent with the filtered sign test (73.9%, $p = 0.006$), confirming no systematic bias from the filter.

C GEE Model Diagnostics

Correlation structure robustness. Table 4 compares exchangeable and independence working correlation structures across all nine judge×axis cells. The two structures yield virtually identical results: $|\Delta\text{QIC}| < 0.001$ and maximum coefficient change <0.02 in every cell, with estimated within-pair correlations $\hat{\alpha} \in [-0.25, +0.11]$. All GEE results reported in the main text are therefore robust to correlation structure specification.

Interaction test. Adding a Style×Originality interaction term to the additive GEE specification yields no significant interactions in any of the nine judge×axis cells (all $p > 0.13$, range 0.14–0.96), confirming that the additive model is the correct specification and that regime classifications are robust to model specification. Table 5 reports interaction terms; none are significant, confirming the additive specification.

Table 4: GEE correlation structure comparison across nine judge×axis cells.

Config	Exch. QIC	Indep. QIC	$\lvert\Delta\text{QIC}\rvert$	Max$\lvert\Delta\text{Coef}\rvert$
GPT-4o form.	—	—	<0.001	<0.02
Qwen3 form.	—	—	<0.001	<0.02
Llama-70B form.	—	—	<0.001	<0.02
GPT-4o verb.	—	—	<0.001	<0.02
Qwen3 verb.	—	—	<0.001	<0.02
Llama-70B verb.	—	—	<0.001	<0.02
GPT-4o reg.	—	—	<0.001	<0.02
Qwen3 reg.	—	—	<0.001	<0.02
Llama-70B reg.	—	—	<0.001	<0.02

Table 5: GEE Style × Originality interaction terms. None are significant ($p > 0.13$), confirming the additive model specification.

Judge	Axis	Interaction OR	95% CI	p
GPT-4o	Formality	0.661	[0.18, 2.38]	.527
GPT-4o	Verbosity	1.269	[0.37, 4.34]	.705
GPT-4o	Register	2.601	[0.74, 9.15]	.136
Qwen3-32B	Formality	0.518	[0.14, 1.89]	.320
Qwen3-32B	Verbosity	0.450	[0.13, 1.62]	.221
Qwen3-32B	Register	0.555	[0.15, 2.08]	.383
Llama-70B	Formality	0.968	[0.25, 3.78]	.963
Llama-70B	Verbosity	1.103	[0.34, 3.58]	.870

95% confidence intervals. Table 6 reports 95% CIs for all GEE odds ratios in Table 3.

Table 6: 95% CIs for GEE conditional decomposition (Table 3).

Config	Style OR [CI]	Orig OR [CI]
Qwen3 verb.	13.10 [5.62, 30.55]	0.49 [0.21, 1.14]
GPT-4o verb.	11.34 [4.93, 26.10]	0.45 [0.20, 1.04]
Llama-70B verb.	8.21 [3.62, 18.65]	0.38 [0.17, 0.86]
Gemini verb.	8.98 [3.68, 21.94]	0.46 [0.20, 1.04]
GPT-4o form.	2.58 [1.41, 4.73]	2.32 [1.27, 4.26]
Qwen3 form.	4.20 [1.99, 8.87]	4.70 [2.22, 9.96]
Qwen3 form.[a]	2.15 [1.46, 3.16]	4.99 [3.40, 7.34]
Llama-70B form.	4.81 [2.48, 9.33]	3.12 [1.61, 6.04]
Gemini form.	1.64 [0.91, 2.98]	2.51 [1.39, 4.54]
GPT-4o reg.	1.88 [1.08, 3.28]	2.21 [1.26, 3.86]
Qwen3 reg.	1.91 [1.01, 3.58]	5.72 [3.02, 10.82]
Llama-70B reg.	3.33 [1.73, 6.42]	3.07 [1.60, 5.92]

[a] $n = 355$; other rows: $n = 88$.

D Rewriter Independence

Evaluator-producer overlap is limited to Qwen3-32B (rewriter = judge); the independence test below bounds this to a 22% Style OR reduction. The other three target judges (GPT-4o, Llama-70B, Gemini) use a different rewriter and are not subject to this confound.

To test whether the Blindspot depends on the rewriter's identity, we replace Qwen3-32B with GPT-4o as rewriter while keeping Qwen3-32B as judge.

Verbosity (Table 7). The Style OR decreases by 22% (13.10 → 10.17), bounding evaluator-producer inflation of the style effect; the pure-style pattern (Originality n.s.) and Blindspot (BT significant, Likert n.s.) both persist.

Table 7: Rewriter independence test: verbosity axis, Qwen3-32B judge. GPT-4o rewriter eliminates evaluator-producer overlap while preserving the Blindspot.

	GPT-4o Rewriter	Qwen3 Rewriter
NLI pass rate	81/100 (81%)	86/100 (86%)
BT P [95% CI]	.735 [.654, .809]	.773 [.703, .843]
Likert ATE (p)	.000 (.838)	∼0 (.179)
Blindspot	Yes	Yes
GEE Style OR (p)	10.17 (<.0001)	13.10 (<.0001)
GEE Orig OR (p)	1.61 (n.s.)	0.49 (n.s.)

Formality (Table 8). GPT-4o rewriter produces formality-manipulated pairs yielding BT 64.4% [56.1, 72.8], statistically indistinguishable from the Qwen3-32B baseline (66.5% [58.5, 73.9]; $\Delta = -2.1$pp). The Likert Blindspot replicates with the independent rewriter ($p = 0.184$, n.s.).

Table 8: Rewriter independence test: formality axis, Qwen3-32B judge.

	GPT-4o Rewriter	Qwen3 Rewriter
NLI pass rate	90/100 (90%)	88/100 (88%)
BT P [95% CI]	.644 [.561, .728]	.665 [.585, .739]
Likert d (p)	−.081 (.184)	.011 (.908)
Blindspot	Yes	Yes

E NLI Threshold Sensitivity Sweep

Table 9 reports BT win probabilities across 11 NLI entailment thresholds for all four judge×axis configurations. All 44 combinations are significant ($p < 0.05$), and within each configuration BT varies by at most 3pp, ruling out threshold selection bias. The stability reflects the bimodal NLI score distribution: most pairs score above 0.95 or below 0.30, so thresholds between 0.50 and 0.95 admit nearly identical pair sets.

Table 9: BT win probability across NLI thresholds. All 44 cells significant ($p < 0.05$). n: pairs passing threshold.

Thresh	Formality BT		Verbosity BT	
	Qwen3	GPT-4o	Qwen3	GPT-4o
.50	.648	.615	.770	.768
.55	.648	.615	.770	.768
.60	.656	.621	.770	.768
.65	.656	.621	.767	.766
.70	.657	.623	.767	.766
.75	.657	.623	.773	.760
.80	.657	.623	.773	.760
.85	.657	.623	.773	.760
.90	.665	.630	.773	.760
.95	.653	.629	.765	.752
.98	.649	.642	.771	.769

F Direction-Aware Gate Validation

For verbosity, the direction-aware NLI gate admits 21 additional pairs (32%) compared to strict bidirectional verification. These pairs show consistent preference direction (BT 73.8% vs. 78.5% for bidirectional-pass pairs), and overall BT estimates are statistically indistinguishable ($\Delta = -1.1$pp, 95% CI $[-5.6, +3.0]$), confirming that direction-aware gating recovers valid pairs without inflating the effect. Direction-aware gating admits 21 additional pairs with consistent bias direction (BT 73.8%, $p = 0.004$); strict bidirectional-only pairs show BT 78.5% ($n = 65$) versus the aggregate BT 77.3% ($n = 86$), a difference of $\Delta = -1.1$pp with the 95% CI including zero. The direction-aware gate does not introduce selection bias. For formality, strict bidirectional verification is appropriate: a relaxed (direction-aware) gate would admit 9 additional pairs with reversed preference (BT 33.4%), indicating content preservation failure rather than genuine style preference. This asymmetry empirically justifies the axis-dependent gating strategy described in Section 3.2.

G Claude Sonnet 4 Equivalence Testing

Claude Sonnet 4 serves as negative control ($n = 87$; BT 56.2%/54.1%, $p = 0.091/0.284$). Formal equivalence testing (TOST; Lakens, 2017) confirms equivalence at $\Delta = 15$pp ($p < 0.05$ for both axes), obtained at approximately 76% power. The mini-

mum equivalence margin achieving 80% power at $n = 87$ is $\Delta = 15.7$pp; at $\Delta = 10$pp TOST is underpowered, and a tighter margin requires larger n (§6). The $\Delta = 15$pp margin bounds any undetected Claude style preference to <15pp—substantively distinct from GPT-4o (13–23pp) and Qwen3-32B (17–27pp), and below the smallest confirmed bias (+13pp for GPT-4o formality).

H Bradley-Terry Goodness of Fit and Stratified Analysis

Bradley-Terry goodness-of-fit tests confirm adequate model fit for both axes (formality: deviance $= 52.80$, df $= 45$, $p = 0.198$; verbosity: $p = 0.905$). Stratified analysis of consistent pairs by original-text direction reveals no preference heterogeneity for verbosity (Fisher $p = 0.453$) but significant direction asymmetry for formality (88.9% vs. 52.6%, Fisher $p = 0.015$), independently confirming the originality confound structure identified by GEE (Table 3).

To assess robustness to originality confounding, we restrict to pairs where the judge assigned equal Likert scores to original and counterfactual texts (eliminating detectable quality differences). Style preference persists: formality BT = 67.3% ($n = 55$, $\Delta = +0.8$pp from full sample, 95% CI $[-11.0, +12.6]$); verbosity BT = 76.0% ($n = 50$, $\Delta = -1.3$pp, 95% CI $[-13.3, +10.3]$). In the anti-originality subset (pairs where the original never wins), effects strengthen (formality 75.0%, verbosity 86.1%), confirming that style preference is not an artifact of originality confounding.

I Bradley-Terry Confidence Intervals

Table 10 reports bootstrap 95% confidence intervals for all Bradley-Terry estimates in Table 1.

J Bootstrap Attenuation Analysis

To distinguish measurement attenuation from insufficient statistical power, we subsample the $n = 355$ formality replication (Qwen3-32B) at seven sample sizes and compute Likert Cohen's d with bootstrap 95% CIs at each (Table 11). Under a power limitation, d would remain constant while only p-values improve with larger n; under measurement attenuation, d itself converges to zero. The data support attenuation: $d_{\text{median}} = -0.017$ across all sample sizes, with CIs narrowing from ± 0.3 to ± 0.1 but never excluding zero. The same data yield BT = 66.5% ($p < 0.0001$), confirming that the pairwise signal is robust. Verbosity shows a similar pattern ($d \approx 0.15$–0.19, stable, CIs always containing zero; BT = 76–77%, $p < 0.0001$). Within the agnostic framework of §5.1, this evidence favors the measurement attenuation interpretation but does not exclude construct divergence.

Table 10: Bootstrap 95% CIs for BT win probabilities (10,000 pair-level resamples).

Bench	Judge	Axis	BT P [95% CI]
LLMBar	GPT-4o	Form.	.630 [.557, .703]
	GPT-4o	Verb.	.760 [.667, .853]
	GPT-4o	Reg.	.570 [.468, .673]
	Qwen3	Form.	.665 [.585, .739]
	Qwen3	Verb.	.773 [.703, .843]
	Qwen3	Reg.	.553 [.479, .628]
	Llama-70B	Form.	.695 [.619, .765]
	Llama-70B	Verb.	.735 [.655, .809]
	Llama-70B	Reg.	.635 [.555, .712]
	Claude	Form.	.562 [.489, .636]
	Claude	Verb.	.541 [.459, .616]
AEval	GPT-4o	Form.	.727 [.663, .785]
	Qwen3	Form.	.657 [.581, .733]
	Qwen3	Verb.	.841 [.774, .902]
MTB	GPT-4o	Form.	.692 [.623, .753]
	Qwen3	Form.	.712 [.630, .788]

Table 11: Bootstrap attenuation analysis: formality × Qwen3-32B. Likert d remains flat near zero across sample sizes while BT is stable at 66.5%.

n	Likert d	95% CI	Contains 0?
50	−0.017	[−0.308, 0.262]	Yes
100	−0.015	[−0.219, 0.176]	Yes
150	−0.019	[−0.179, 0.145]	Yes
200	−0.016	[−0.157, 0.119]	Yes
250	−0.018	[−0.144, 0.107]	Yes
300	−0.018	[−0.131, 0.095]	Yes
355	−0.017	[−0.123, 0.087]	Yes

K Formal Power Analysis

Table 12 reports the sample size required for Likert ATE to achieve 80% power at observed effect sizes across all six judge×axis cells. Required sample sizes range from 336 (Qwen3-32B verbosity, $d = 0.153$) to 27,159 (Qwen3-32B formality, $d = -0.017$), representing 4–206× the current n. For five of six cells, the required n exceeds 1,200—far beyond typical evaluation budgets—converging with the bootstrap attenuation curve (Appendix J) to confirm the Blindspot as a structural measurement limitation.

Table 12: Required Likert sample size for 80% power at observed d. BT detects all six effects at $p < 0.001$ with $n \leq 100$.

Judge	Axis	Observed d	Required n	Ratio
GPT-4o	Form.	0.073	1,473	16.7×
GPT-4o	Verb.	0.080	1,227	14.3×
Qwen3	Form.	−0.017	27,159	76.5×
Qwen3	Verb.	0.153	336	3.9×
Llama-70B	Form.	0.021	18,157	206.3×
Llama-70B	Verb.	0.034	6,737	78.3×

L Verbosity Sensitivity Hierarchy

Table 13 replicates the seven-method sensitivity analysis (Table 2) on the verbosity axis ($n = 86$ NLI-filtered pairs, Qwen3-32B judge, LLMBar). The attenuation pattern is more extreme than formality: all six Likert-based methods remain non-significant (lowest $p = 0.213$) despite a stronger pairwise signal (BT 77.3% vs. 66.5% for formality), and Likert effect sizes ($d = 0.036$–0.056) are near zero.

Table 13: Sensitivity hierarchy for verbosity ($n = 86$, Qwen3-32B, LLMBar). All Likert methods fail; only BT reaches significance.

Method	Statistic	p	Sig?
BT model	77.3% [70.3, 84.3]	<0.001	Yes
Likert ATE	$d = 0.056$	0.691	No
Binarized χ^2	OR = 2.16	0.213	No
Ordinal reg.	$\beta = 0.159$	0.575	No
Paired t-test	$d = 0.056$	0.607	No
Mann-Whitney U	$r = -0.046$	0.578	No
Cohen's d	$d = 0.036$	—	No

Effect sizes use style-aligned scoring (verbose − concise), isolating style preference direction. Table 1 reports RATE-compatible ATE (counterfactual − original), which conflates style and originality effects.

M 100-Point Continuous Likert Scale

To test whether the Blindspot reflects integer-scale granularity rather than a paradigm difference between absolute and relative evaluation, we replicate the formality and verbosity experiments using a 100-point continuous scale (0–100) with Qwen3-32B as judge (Table 14). Both axes remain non-significant: formality $p = 0.725$ (tie rate 65.9%) and verbosity $p = 0.117$ (tie rate 51.2%), while the same data yield BT 66.5% ($p < 0.0001$) and 77.3% ($p < 0.0001$) respectively. The persistence of high tie rates even with 101 possible values confirms that LLM judges do not exploit finer granularity to express style preferences; the Blindspot arises from absolute-vs-relative evaluation paradigms, not scale resolution.

Table 14: 100-point continuous Likert vs. pairwise (BT). The Blindspot persists despite 101 possible score values.

Axis	Likert p	Tie rate	BT P	BT p
Formality	.725	65.9%	.665	<.0001
Verbosity	.117	51.2%	.773	<.0001

N NLI Gate Ablation

Table 15 compares BT estimates across NLI-pass, NLI-fail, and all pairs. For formality, NLI-fail pairs reverse to 41.7% (preferring casual over formal), demonstrating that the gate removes content-confounded pairs rather than amplifying style signal. The small NLI-fail sample ($n \approx 12$) limits precision; this serves as a bounding analysis. For verbosity, NLI-fail pairs remain at 75.0%, consistent with the dominant style effect (OR = 13.10) overwhelming content noise.

Table 15: NLI gate ablation: BT win probability by NLI subset.

Axis	NLI-pass	NLI-fail	All
Formality	66.5%	41.7%	63.0%
Verbosity	77.3%	75.0%	76.0%

O Rewriting Artifact Control

Same-style double-rewrite (DR) controls quantify the rewriting artifact using BT models (consistent with all other pairwise analyses). Table 16 reports BT $P(\text{original})$—the probability the judge prefers the original over its same-style DR. Both judges systematically prefer LLM-polished text across axes; for verbosity, BT $P(\text{original}) < 0.17$ indicates that rewriting quality dominates when content is held constant, validating the GEE decomposition that separates style from artifact effects. Same-style double-rewrite controls for style-neutral polishing artifacts but cannot exclude style-specific polishing interactions (e.g., formal rewriting may polish differently than casual rewriting).

P Implementation Details

Models and APIs. Style-controlled rewriting uses Qwen3-32B (temperature 0.7, top-p 0.9). Target judges: GPT-4o (`gpt-4o-2024-08-06`, accessed April–May 2026), Qwen3-32B (locally hosted), Llama-3.3-70B (locally hosted

Table 16: Rewriting artifact: BT $P(\text{orig})$ vs. same-style double-rewrite (bootstrap 95% CI, 10,000 resamples). Values <0.5: judges prefer LLM-polished text.

Axis	Judge	BT $P(\text{orig})$ [CI]	p
Form.	GPT-4o	.418 [.342, .500]	.037
	Qwen3	.377 [.303, .451]	.0006
Verb.	GPT-4o	.167 [.084, .262]	$<.0001$
	Qwen3	.082 [.035, .140]	$<.0001$

Table 17: Holm-Bonferroni sensitivity analysis: 8-test family (4 judges $\times$ 2 primary axes). All results remain significant under the expanded family ($\alpha = 0.05$).

Rank	Judge	Axis	BT P	Raw p	Threshold	Adj. p
1	Gemini Flash	Verb.	.708	$<.0001$	.00625	$<.001$
2	GPT-4o	Verb.	.760	.0001	.00714	.0007
3	Qwen3-32B	Form.	.665	.0001	.00833	.0007
4	Qwen3-32B	Verb.	.773	.0001	.0100	.0007
5	Llama-70B	Form.	.695	.0001	.0125	.0007
6	Llama-70B	Verb.	.735	.0001	.0167	.0007
7	GPT-4o	Form.	.630	.0004	.0250	.0008
8	Gemini Flash	Form.	.574	.0306	.0500	.0306

via vLLM). Negative control: Claude Sonnet 4 (`claude-sonnet-4-20250514`, accessed May 2026). All judge calls use temperature 0 and `max_tokens` $= 2048$.

NLI gate. Content preservation is verified using DeBERTa-v3-large (He et al., 2023) fine-tuned on MNLI, with entailment threshold $\tau = 0.90$. Bidirectional entailment is applied for formality and register; direction-aware entailment for verbosity (Section 3.2).

Statistical methods. Bradley-Terry parameters are estimated via spectral initialization followed by minorization-maximization (Hunter, 2004) using the `choix` Python library (default regularization). All bootstrap CIs use 10,000 pair-level resamples with seed 42. GEE models use exchangeable working correlation with robust (sandwich) standard errors via `statsmodels` (Section 5.2; diagnostics in Appendix C).

Benchmarks. LLMBar (Zeng et al., 2024): 100 instruction-response pairs ($n = 88$ after NLI gate, $n = 355$ for scaled replication). AlpacaEval (Dubois et al., 2025): 100 sampled pairs. MT-Bench (Zheng et al., 2023): 80 multi-turn pairs.

Code and data. All code, prompts, and experimental data will be released upon acceptance under an open-source license.

Q Holm-Bonferroni Sensitivity Analysis

Table 17 reports adjusted p-values under the full 8-test family (4 judges $\times$ 2 primary axes), including Gemini-2.5-Flash which is excluded from the primary 6-test family (§4.1). All results remain significant at $\alpha = 0.05$.

R Cross-Axis Contamination Analysis

Our per-axis design assumes that rewriting along one style axis does not substantially alter the text along other axes. We quantify cross-axis contamination by measuring off-target surface proxies before and after rewriting. For formality rewrites ($n = 355$), we measure word count and sentence count as verbosity proxies; for verbosity rewrites ($n = 86$), we measure type-token ratio (TTR) as a lexical diversity proxy for formality, and Flesch-Kincaid (FK) grade level.

Table 18: Cross-axis contamination: effect of rewriting along one style axis on surface proxies of the other axis. Cohen's d magnitudes below 0.20 indicate negligible contamination (Cohen, 1988).

Direction	Proxy	Δ	d	Interp.
Form. $\rightarrow$ Verb.	Word count	-3.4%	-0.23	Small
Form. $\rightarrow$ Verb.	Sent. count	-0.2%	-0.03	Negligible
Verb. $\rightarrow$ Form.	TTR	-0.006	-0.04	Negligible
Verb. $\rightarrow$ Form.	FK grade	—	0.50	Confounded*

* FK $= f$(sentence length, word length); sentence length change is inherent to verbosity manipulation, not formality contamination.

Formality rewrites leave sentence structure virtually unchanged ($d = -0.03$); the small word count shift ($d = -0.23$) reflects word-level adjustments (contractions, elaboration) that are directionally symmetric (formal$\rightarrow$casual -5.0%, casual$\rightarrow$formal $+5.6\%$) and do not constitute verbosity contamination. Verbosity rewrites leave lexical diversity unchanged (TTR $d = -0.04$); the elevated FK grade ($d = 0.50$) is an artifact of FK's dependence on sentence length, which changes by design in verbosity manipulation. These results confirm that cross-axis contamination is small to negligible for the proxies measured; the GEE originality term (§5.2) provides additional protection by absorbing any residual quality confound.

S Human Annotation Agreement Statistics

Two independent annotators evaluated all 50 NLI-scored pairs for content preservation. Annotator 1 (A1) judged 42/50 pairs content-preserving; Anno-

tator 2 (A2) judged 33/50, reflecting systematically stricter criteria. All inter-annotator disagreements are predominantly unidirectional (A1-Yes/A2-No: 10; A1-No/A2-Yes: 1), confirming a calibration difference rather than random noise. Inter-annotator agreement: raw 78% (39/50), Cohen's $\kappa = 0.438$ (moderate), Gwet's AC1 $= 0.648$ (substantial). The κ–AC1 gap reflects prevalence skew (Feinstein and Cicchetti, 1990): the high base rate of content-preserving pairs inflates chance agreement, compressing κ.

Among the 25 gate-passed pairs, A1 confirms all 25 as content-preserving (precision 100%; Wilson CI $[86.7\%, 100\%]$); A2 confirms 21/25 (84%; CI $[65.3\%, 93.6\%]$). The four A2 disagreements (S003, S018, S039, S049) are all verbosity-axis condensation pairs where A2 applied stricter meaning-equivalence criteria to information-reducing rewrites. Under majority vote, pass precision is 100% (25/25).

NLI–annotator agreement: NLI–A1 $\kappa = 0.320$, NLI–A2 $\kappa = 0.360$. Both κ values are nominally "fair" but deflated by the same prevalence mechanism. The clinically relevant statistic for a precision-oriented gate is not overall agreement but *pass precision*: at 84–100% observed pass precision, the gate's validity for its intended function—blocking content contamination from the analysis—is confirmed under both lenient and strict annotation standards. All A1 disagreements with the NLI gate occur on gate-*rejected* pairs; A2's four gate-passed disagreements reflect subjective meaning boundaries on verbosity condensation rather than content-quality contamination.

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